

Total No. of Questions : 8]

SEAT No. :

PD4492

[Total No. of Pages : 2

[6403] 300

T.E. (Computer Engineering) (Honors)

VIRTUAL REALITY AND AUGMENTED REALITY

Augmented Reality

(2019 Pattern) (Semester - VI) (310703)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicates full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Explain natural feature tracking by detection in augmented reality. [9]
b) How pose estimation from Homography is done in marker tracking. Explain. [8]

OR

- Q2)** a) Explain incremental tracking in details. [9]
b) What are multiple cameras infrared tracking in AR? [8]

- Q3)** a) Explain Marker detection procedure. [6]
b) Write note on : [6]
i) 2D Barcode marker
ii) Imperceptible markers
c) Explain the term Hybrid tracking in detail. [6]

OR

- Q4)** a) Explain Visual tracking in AR. [6]
b) When to use marker-based tracking and how to speed up marker detection in detail? [6]
c) How to select a marker type. Explain with respect to system requirement. [6]

- Q5)** a) Write note on : [8]
i) Scene generator
ii) Projection display
b) Explain virtual retinal system in detail. [9]

OR

P.T.O.

Q6) a) Explain optical see-through and video see-through head mounted display. [8]

b) Explain with diagram monitor based augmented reality display. [9]

Q7) a) What is mixed reality? Explain the different application of mixed reality. [9]

b) Explain parallel tracking and mapping (PTAM) in detail. [9]

OR

Q8) a) Explain in detail working principle of SLAM. [9]

b) Explain Semi-direct monocular Visual Odometer (SVO) in detail. [9]

